Jiahua Zhang

Incoming MEng in Electrical & Computer Engineering, National University of Singapore · Jan 2027 – 2028 (expected)

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EDUCATION

National University of Singapore Jan 2027 – 2028 (expected)

MEng, Department of Electrical and Computer Engineering · Supervised by Prof. Changsheng Wu

Nanyang Technological University, Singapore Aug 2022 – Jun 2026

B.Comp. (Honours) in Computer Science, College of Computing and Data Science

- Specialisation in Data Science.
- Five papers published as an undergraduate, including a first-authored CCF-A paper; research carried out at a national key laboratory and several universities.

PUBLICATIONS

1. **Jiahua Zhang**, Y. Tian, J. Liang, D. Lin, Y. Cai, Y. Liu, Z. Huang, J. Ni, J. He. Process Reward Model as Cross-Task Learner: An Extension of Vision-Language Model to Zero-Shot Medical Reasoning. *Proceedings of the 34th ACM International Conference on Multimedia (ACM MM)*, 2026.
2. **Jiahua Zhang**, J. Liang, Y. Cai, J. Ni, J. He. Can LLM Understand Medical Imaging with Long Context? *IEEE International Conference on Multimedia and Expo (ICME)*, 2026. ORAL
3. **Jiahua Zhang**, Y. Tian. DiffiT-HSFDA: Diffusion Based Source-Free Domain Adaptation for Histopathology. *Advanced Intelligent Computing Technology and Applications (ICIC)*, Springer LNCS 15867, pp. 301–311, 2025. ORAL
4. C. Zhang, X. Wang, H. Zhang, **Jiahua Zhang**, H. Zhang, C. Liu, P. Han. LayerLog: Log Sequence Anomaly Detection Based on Hierarchical Semantics. *Applied Soft Computing*, 132:109860, 2023. (IF 7.8, JCR Q1)
5. C. Zhang, X. Wang, **Jiahua Zhang**, S. Li, H. Zhang, C. Liu, P. Han. VESC: A New Variational Autoencoder Based Model for Anomaly Detection. *International Journal of Machine Learning and Cybernetics*, 14(3):683–696, 2022. (IF 2.9)

RESEARCH EXPERIENCE

Nanyang Technological University — CAO Exchange Programme, Shenzhen Nov 2025 – Feb 2026

Research Intern — BioMedPRM: Process Reward Modelling for Biomedical Reasoning

- Supervised each step of an LLM reasoning chain rather than only its final answer, improving the reliability of reasoning in biomedical question answering.
- Designed and evaluated a process reward model that transfers across tasks, with emphasis on robustness under distribution shift and in out-of-distribution clinical settings.
- **Output:** one first-authored paper at ACM Multimedia 2026.

Guangzhou Medical University — State Key Laboratory of Respiratory Disease Feb 2024 – Present

Research Collaborator, Thoracic Surgery / Thoracic Oncology Group

- Worked with Prof. Jianxing He's thoracic surgery team on a visual foundation model programme for lung-cancer diagnosis from histopathology and CT.
- **Source-free domain adaptation:** found that weakly supervised segmentation of lung-cancer histopathology loses region-of-interest activation when transferred across centres with different staining protocols; designed a diffusion-based source-free adaptation pipeline that restores high-accuracy segmentation without source data or extra annotation.
- **Training-free multimodal LLM:** built a training-free framework for CT and whole-slide images, adapting long-context models to high-dimensional medical imaging without task-specific fine-tuning.
- **Output:** first-authored orals at ICIC 2025 and IEEE ICME 2026.

Nanyang Technological University — School of Social Sciences (Psychology) Oct 2024 – May 2025

Research Assistant, supervised by Asst. Prof. Luo Lizhu

- Represented high-dimensional biomedical data — functional MRI, whole-slide images — as graphs so that language models can reason over far more variables than a context window admits.
- Proposed a training-free graph in-context learning method for medical multimodal prediction and benchmarked it against fine-tuned baselines.

Pohang University of Science and Technology (POSTECH) — DPNM Laboratory Jun 2024 – Aug 2024

POSTECH Summer Program (PSP), supervised by Prof. James Won-Ki Hong

- Designed a convolutional diffusion-denoising pipeline for MRI and X-ray segmentation and spinal disease classification, improving results on the BUU and lumbar-spine MRI datasets.
- Applied representation learning and NLP to hospital chief-complaint text, evaluating TF-IDF and other vectorisation and clustering methods, and developed lightweight algorithms for edge computing.

Nanyang Technological University — S-Lab for Advanced Intelligence Mar 2024 – Jun 2024

Student Research Assistant — LLM-based video generation

- Developed and applied noise-detection algorithms to improve the visual quality of AI-generated video.
- Built the label preprocessing pipeline for a generated-video quality evaluation benchmark.

Harbin Institute of Technology, Shenzhen Jan 2022 – Jul 2022

Research Assistant

- Owned data processing for two anomaly-detection studies: cleaning and preprocessing large-scale log and time-series data, and preparing training and evaluation sets.
- Produced the results visualisations for both studies, including the comparative and ablation figures reported in the papers.
- **Output:** two co-authored journal papers.

ENGINEERING EXPERIENCE

Guangzhou Center for Disease Control / National Key Laboratory May 2026 – Oct 2026

Research Engineer

- **Reasoning-chain specification (laboratory):** defined the rules for medical chain-of-thought — how EHR reasoning steps are decomposed, how clinical visual evidence is marked, and the RLHF annotation principles — as the single reference for multimodal data production and model evaluation.
- **Evaluation framework (laboratory):** designed the reasoning-chain evaluation framework, defining dimensions, scoring rules and spot-check procedures used to compare models and data versions across iterations.
- **Annotation operations (laboratory):** decomposed data-cleaning requirements into executable rules and instructions for a team of medical students; ran training, periodic spot checks and feedback loops to hold quality and delivery schedule.
- **Platform operations (CDC):** ran day-to-day operations of the Xingguang platform A800 cluster, A100 multi-GPU servers and the data platform; gathered compute demand from each team and set allocation and scheduling across several clinical projects.
- **Environment and compliance (CDC):** standardised training and inference environments; maintained storage planning and access control on the data platform, implementing de-identification and audit logging to meet data-security requirements.
- Wrote the rule documents, evaluation guides and platform handbooks used to onboard team members and medical students.

Nanyang Technological University Dec 2024 – Jun 2025

Student Assistant Engineer — HPC

- Operated the school's HPC resources across several GPU generations (V100, T4, A100, A6000): system updates, driver upgrades and fault diagnosis, coordinating repairs and expansion with university IT and hardware vendors.
- Allocated GPU resources to psychology research groups in the School of Social Sciences, matching card types to workload, reviewing and helping shape each group's compute requests, and managing accounts, permissions and storage quotas.
- Acted as the interface between researchers and the platform, handling tickets on environment setup, job submission and performance, and running onboarding sessions for new users.
- Wrote cluster usage guides and FAQ documentation, pushing a standard environment setup that cut the support load.

Shenzhen Wanqi Industrial Co., Ltd. Dec 2023 – Feb 2024

Full-Stack Development Intern

- Worked within the development team on iterative development of .NET / MVC modules of an enterprise ERP system, delivering front- and back-end features with self-testing and integration testing.

SELECTED PROJECTS

Multi-Character Consistent AIGC Agent System 2025 – Present

Core developer · Under review at ACL ARR

- **Problem:** using several character LoRAs at once causes identity bleeding, attribute leakage or dropped characters. Trained a conflict-aware operator offline that rewrites each character's LoRA in parameter space, so any combination is plug-and-play with no inference-time overhead.
- **Production pipeline:** an orchestrator plus message queue dispatches story, storyboard, illustration and localisation tasks; a model-scheduling layer centralises retries and fallbacks, output validation and metric logging, making every generation traceable.
- **Result:** evaluated on five dimensions; with five characters in one frame, identity similarity is 30.3% above the best baseline.

Open-Source Speech Synthesis Toolkit & Hosted Inference Service 2022 – 2023

Independent developer

- Trained and released multi-speaker speech synthesis models (VITS / Bert-VITS2) covering a large voice cast.
- Released vits with chatgpt-gpt3, an open-source TTS API project chaining VITS synthesis to LLM dialogue so characters can both converse and speak; adopted by content creators for video production and livestreaming.
- Deployed and maintained a public inference Space on Hugging Face containers, tuned for latency, cost and concurrency under a constrained compute budget.

SKILLS

Compute & infrastructure: Linux GPU server and cluster operations; NVIDIA driver and CUDA environment management; multi-GPU training and inference deployment; environment standardisation; Shell automation; resource monitoring and fault diagnosis. Hands-on with A800, A100, A6000, V100, T4.

LLMs & reward modelling: process reward models (step-level scoring), vision-language models, long-context multimodal reasoning, test-time computation, training-free in-context learning; benchmark and label construction, multi-dimensional generation-quality evaluation.

Generative models: diffusion models (FLUX / SDXL), LoRA composition and parameter-space analysis (SVD, polar decomposition), domain adaptation and transfer learning, VITS / Bert-VITS2 speech synthesis.

Agents & systems: multi-agent workflow orchestration, message queues and microservices, model-scheduling layers (retry and fallback, output validation, metric logging), online inference deployment and cost optimisation.

ML frameworks: PyTorch, Hugging Face Transformers, PyTorch Geometric, torchvision, torchaudio; large-model training and inference on multi-GPU clusters.

Software & web: Python, C# / .NET, ABP, Flask, Jinja2, Vue, React, MySQL, Android, Unity.

Data engineering: data acquisition and scraping; cleaning and preprocessing pipelines for large-scale clinical, imaging, log and time-series data.

Medical & multimodal data: CT, MRI, X-ray, whole-slide pathology images, functional MRI.

Languages: Mandarin Chinese (native), English (IELTS 7.0).

SERVICE & CAMPUS ACTIVITIES

NTU Chinese Society — Spring Festival Gala, Committee Member 2025

Helped organise the stage production for the NTU 2025 Spring Festival Gala, covering logistics and on-stage support.

NTU Chinese Society — Publicity Team, Committee Member 2023 – 2024

Event manager in the publicity team; wrote and typeset articles for the society's WeChat account across the academic year.

NTU IET Club — Event Manager Aug 2022 – May 2023

Organised IT-themed events and technical competitions for club members and the wider student body; led a committee of five across three events, owning planning, publicity and delivery.

Shenzhen International High School Students Forum — Student Volunteer 2019

Supported delegates from different school systems in preparing and presenting their work, and assisted with on-site coordination.